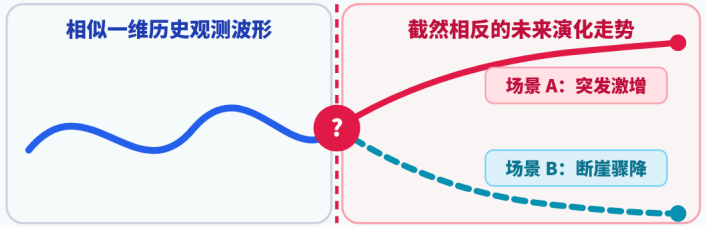
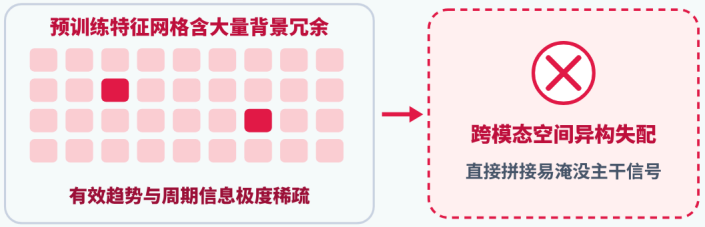
Aurora: 时间预测面临的多模态不足与本文贡献

不足 1 仅时间序列信息不充分



不足 2 现有LLM 面向任务不同,不宜直接用



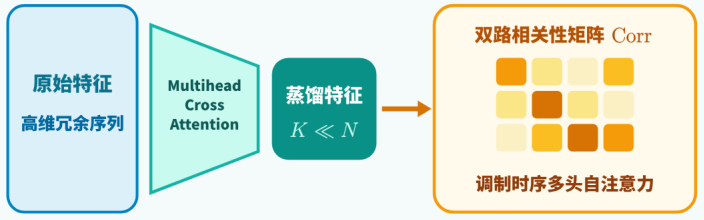
不足 3 传统只能点预测,基于噪声的流匹配生成路径复杂



贡献 1 内生一维与二维双视角及外生文本三路协同编码



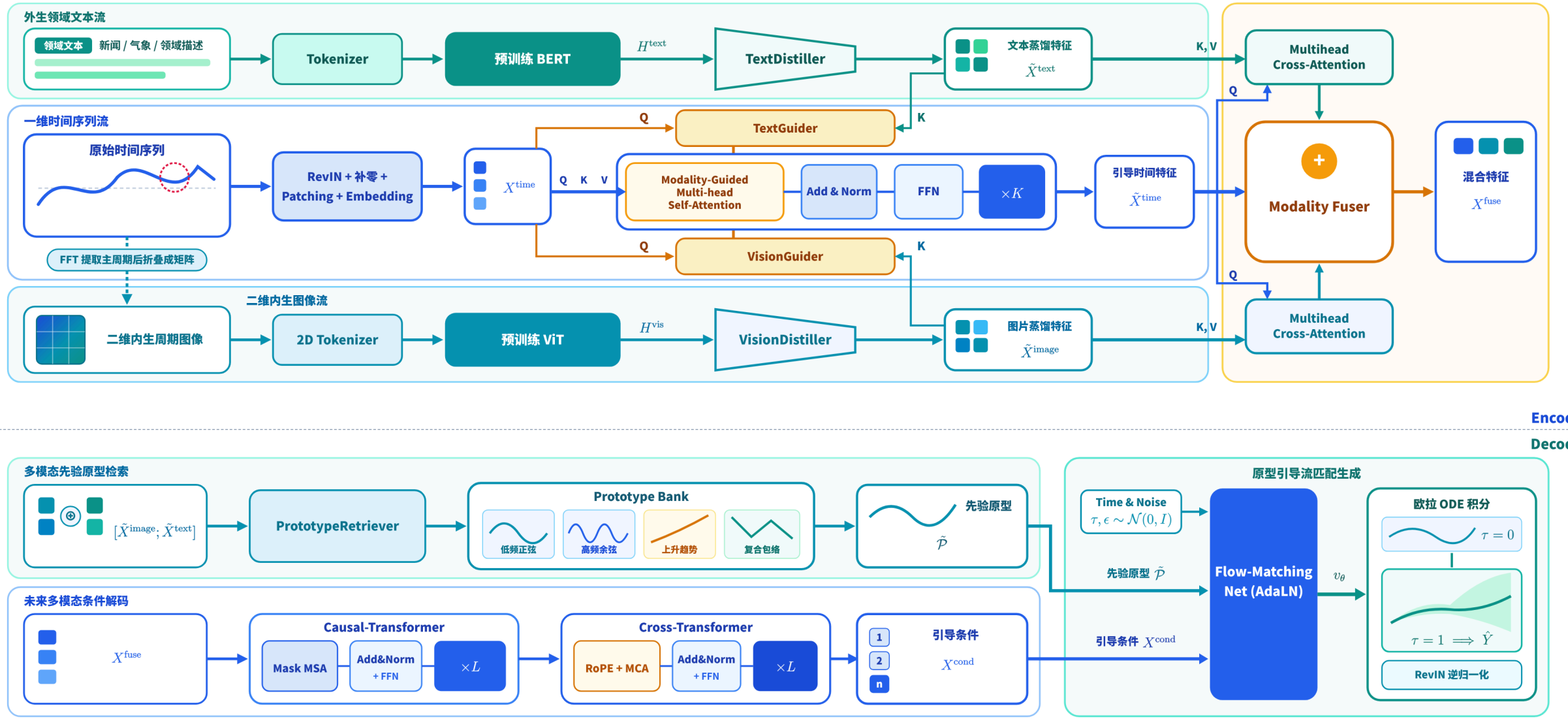
贡献 2 蒸馏与双路引导注意力



贡献 3 基于先验库的流匹配生成

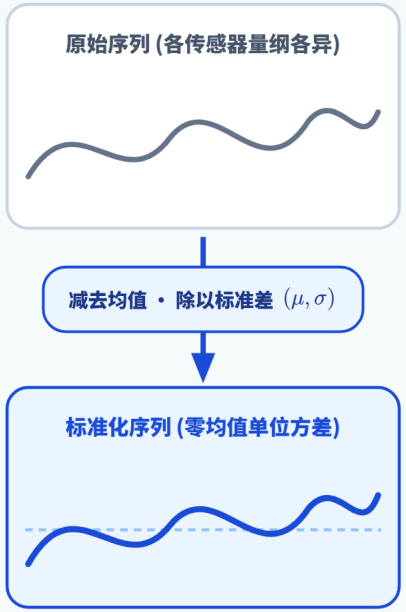


Aurora 算法整体多路架构



时序一维分块与编码

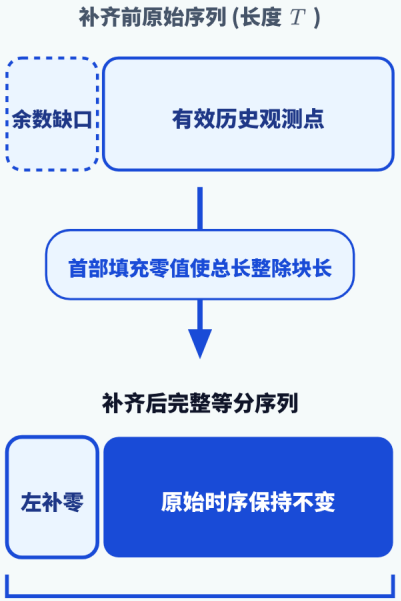
STEP 1 可逆实例归一化



$$\mu = \mathrm{Mean}(x_{1:T}),\ \sigma = \mathrm{Std}(x_{1:T})$$
$$\tilde{x}_t = (x_t - \mu)/(\sigma + \epsilon)$$

$\mathbb{R}^T \to \mathbb{R}^T$

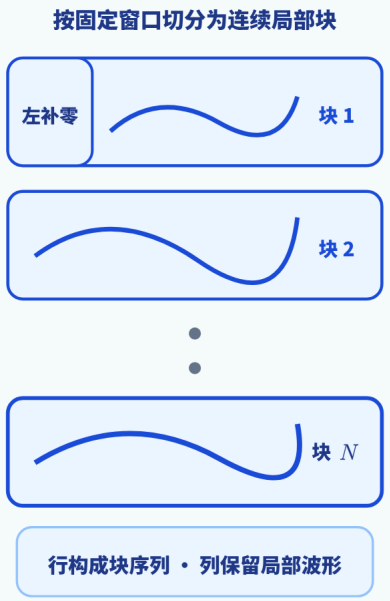
STEP 2 序列首部左补零对齐



$$r = (P - (T \bmod P)) \bmod P$$
$$\tilde{x}_{\mathrm{pad}} = [\mathbf{0}_r,\ \tilde{x}_{1:T}]$$

$\mathbb{R}^T \to \mathbb{R}^{NP}$

STEP 3 非重叠等长窗口分块



$$N = (T + r)/P$$
$$X_p = \mathrm{Reshape}(\tilde{x}_{\mathrm{pad}},\ N,\ P)$$

$\mathbb{R}^{NP} \to \mathbb{R}^{N \times P}$

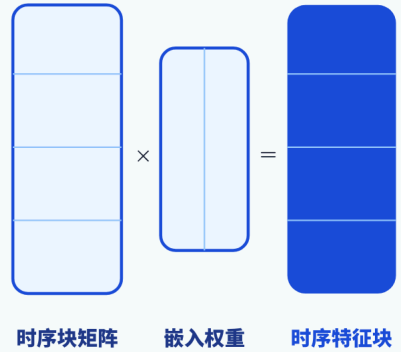
STEP 4 嵌入权重线性插值



$$\alpha(p) = (p/P) \cdot P_0$$
$$W_{\mathrm{time}} = \mathrm{Interp}_{1\mathrm{D}}(W_0,\ P_0 \to P)$$

$\mathbb{R}^{P_0 \times D} \to \mathbb{R}^{P \times D}$

STEP 5 线性映射生成时序特征



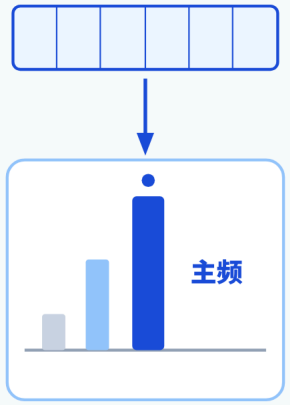
$$X_p \in \mathbb{R}^{N \times P},\ W_{\mathrm{time}} \in \mathbb{R}^{P \times D}$$
$$X^{\mathrm{time}} = X_p\, W_{\mathrm{time}}$$

$\mathbb{R}^{N \times P} \to \mathbb{R}^{N \times D}$

时序的二维内生图像构建与编码

STEP 1 频域提取主导周期

时序网格



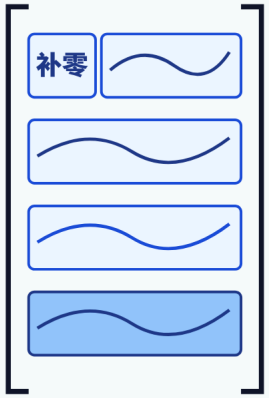
周期 P

$$f^* = \arg\max |\mathrm{FFT}(\tilde{x})|$$
$$P = \lfloor T/f^* \rfloor$$

$\mathbb{R}^T \to P \in \mathbb{Z}^+$

STEP 2 二维周期矩阵折叠

周期折叠



横轴：周期相位

纵轴：相同相位

$$K_p = \lceil T/P \rceil$$
$$G = \mathrm{Fold}_{2D}(\tilde{x}_{\mathrm{pad}}, K_p, P)$$

$\mathbb{R}^{K_p P} \to \mathbb{R}^{K_p \times P}$

STEP 3 双线性插值放缩

双线性插值



尺寸 $H \times W$

适配视觉模型

$$G \in \mathbb{R}^{K_p \times P}$$
$$I_1 = \mathrm{Resize}(G, H, W)$$

$\mathbb{R}^{K_p \times P} \to \mathbb{R}^{H \times W}$

STEP 4 三通道复制扩展

通道复制



复制三份

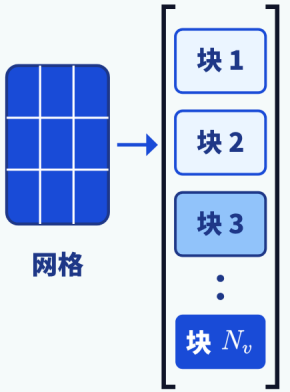
适配视觉模型

$$I_1 \in \mathbb{R}^{H \times W \times 1}$$
$$I_3 = \mathrm{Repeat}(I_1,\ 3)$$

$\mathbb{R}^{H \times W} \to \mathbb{R}^{H \times W \times 3}$

STEP 5 二维网格切块展平

图像切块



每块三通道

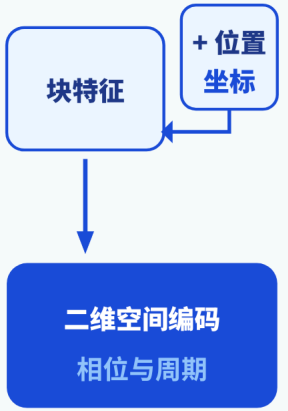
展平为向量

$$N_v = (H/S) \cdot (W/S)$$
$$U_v = \mathrm{Patch}(I_3,\ S^2)$$

$\mathbb{R}^{H \times W \times 3} \to \mathbb{R}^{N_v \times 3S^2}$

STEP 6 线性投影与位置编码

位置编码

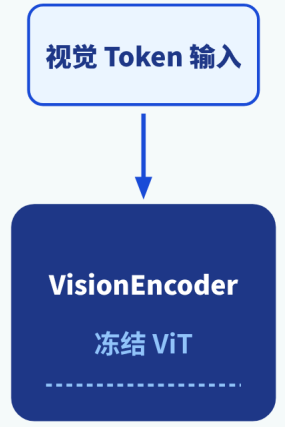


$Z_v^{(0)} \in \mathbb{R}^{N_v \times D}$

$$U_v W_v \in \mathbb{R}^{N_v \times D}$$
$$Z_v^{(0)} = U_v W_v + E_{\mathrm{pos}}^{2D}$$

$\mathbb{R}^{N_v \times 3S^2} \to \mathbb{R}^{N_v \times D}$

STEP 7 冻结 ViT 视觉编码



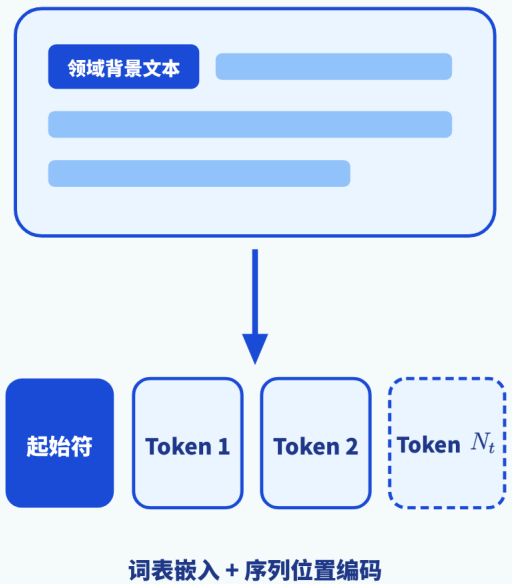
$H^{\mathrm{vis}} \in \mathbb{R}^{N_v \times D}$

$$Z_v^{(0)} \in \mathbb{R}^{N_v \times D}$$
$$H^{\mathrm{vis}} = \mathrm{ViT}(Z_v^{(0)})$$

$\mathbb{R}^{N_v \times D} \to \mathbb{R}^{N_v \times D}$

文本编码与多模态蒸馏

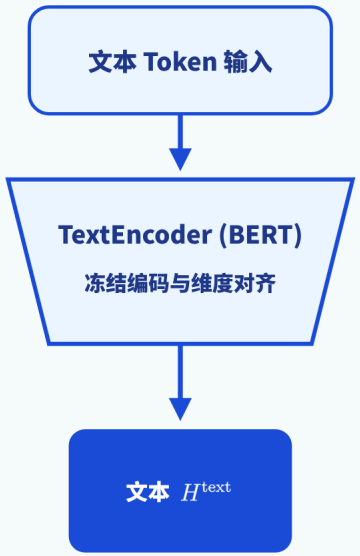
STEP 1 领域文本分词与位置嵌入



$$U_t = \text{Tokenizer}(\text{Text})$$
$$Z_t^{(0)} = U_t W_e + E_{\text{pos}}^{\text{text}}$$

$\text{Text} \to \mathbb{R}^{N_t \times D}$

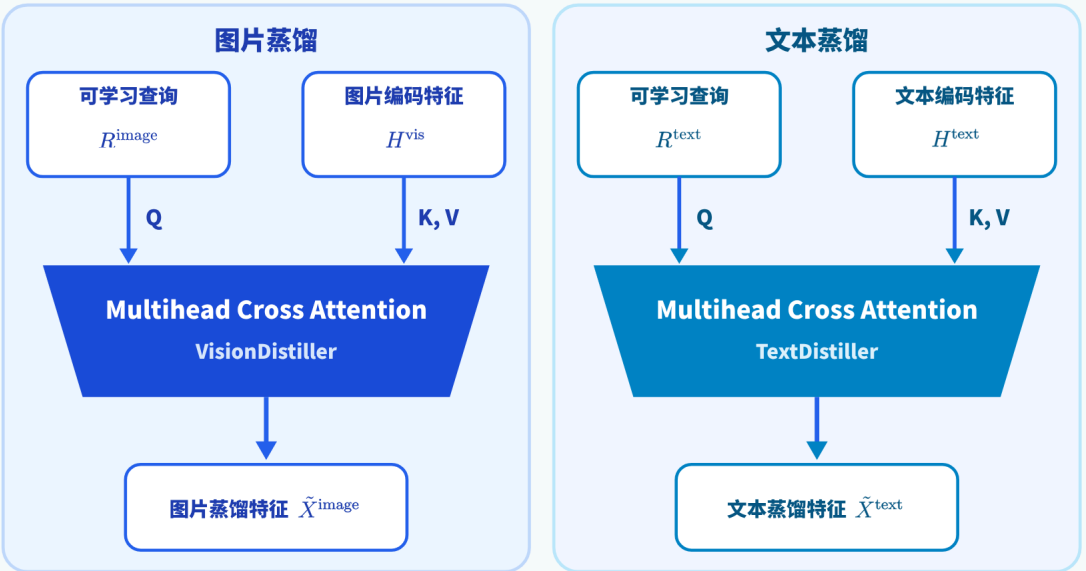
STEP 2 冻结预训练文本编码



$$H^{\text{text}} = \text{BERT}(Z_t^{(0)})$$

$\text{Text} \to \mathbb{R}^{N_t \times D}$

STEP 3 双路交叉注意力蒸馏

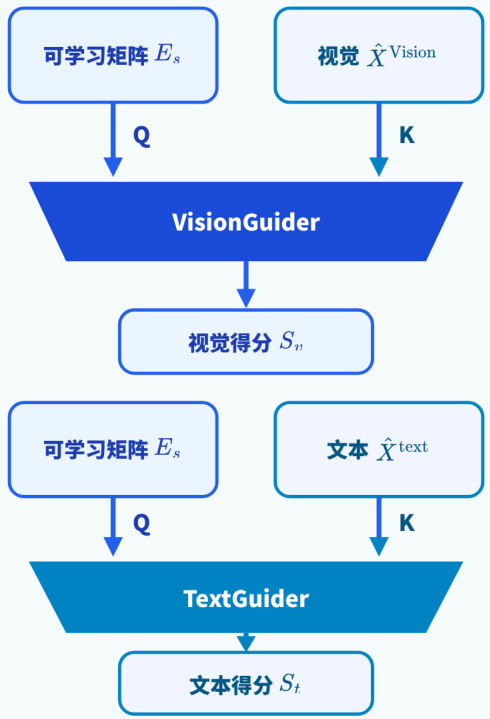


$$\tilde{X}^{\text{image}} = \text{VisionDistill}(R^{\text{image}}, H^{\text{vis}})$$
$$\tilde{X}^{\text{text}} = \text{TextDistill}(R^{\text{text}}, H^{\text{text}})$$

$\mathbb{R}^{N_{\text{raw}} \times D} \to \mathbb{R}^{K \times D}$

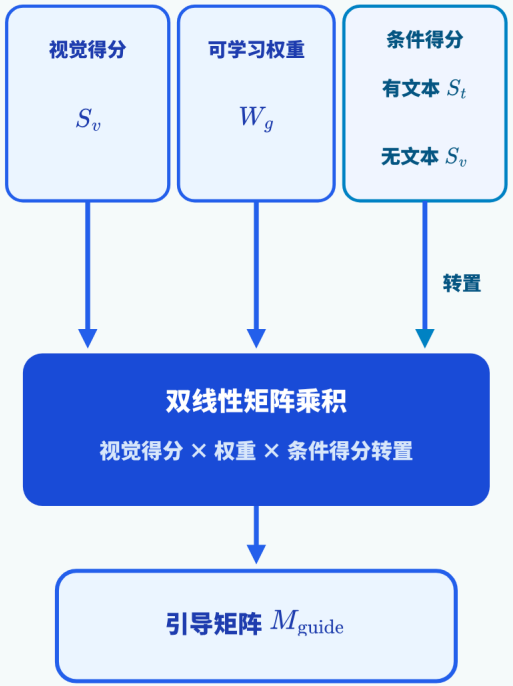
双路引导的自注意力与特征融合

STEP 1 引导器生成模态得分



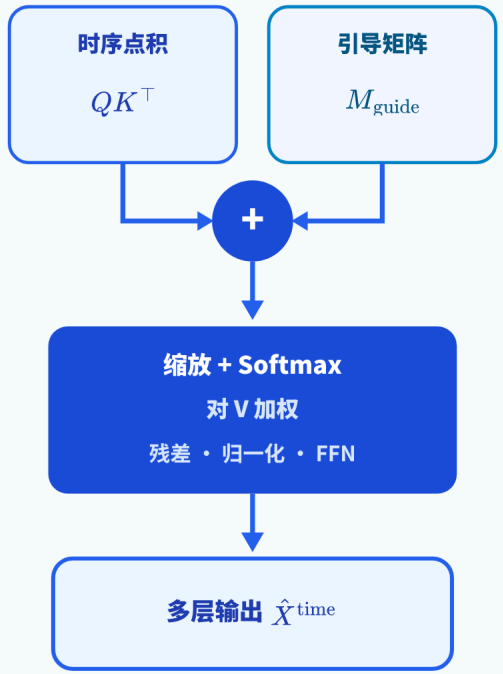
$$Q^m = E_s W^Q_{s\to m}, \quad K^m = \hat{X}^m W_m$$
$$S_m = Q^m (K^m)^\top / \sqrt{d}, \quad m \in \{v, t\}$$

STEP 2 视觉与文本得分合成引导矩阵



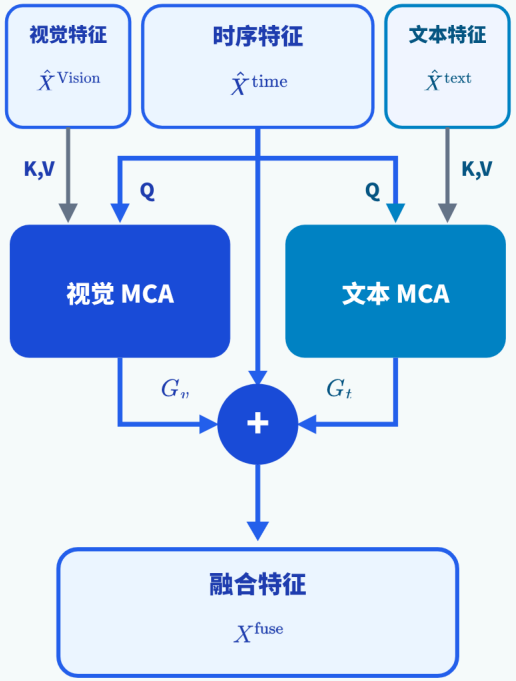
$$M_{\text{guide}} = S_v W_g S_{\text{text}}^\top$$
$$M_{\text{guide}} = S_v W_g S_v^\top \quad \text{(无文本)}$$

STEP 3 引导矩阵加到注意力得分



$$A = \text{Softmax}((QK^\top + M_{\text{guide}})/\sqrt{d})$$
$$\hat{X}^{\text{time}} = \text{Encoder}(X^{\text{time}}; M_{\text{guide}})$$

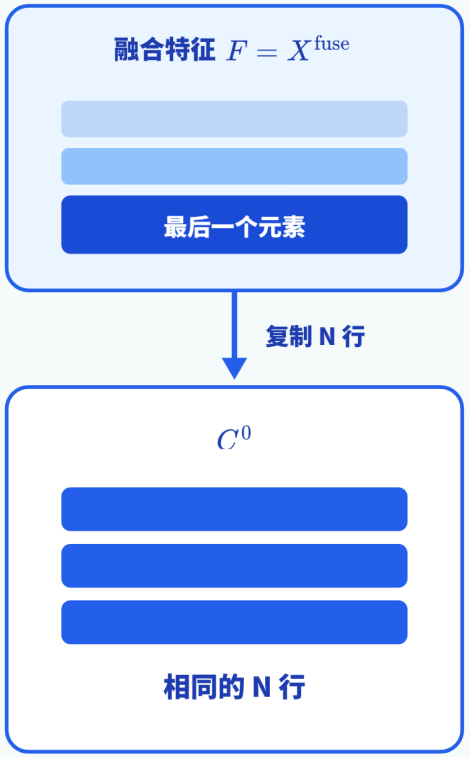
STEP 4 多模态特征融合



$$G_v = \text{MCA}(\hat{X}^{\text{time}}, \hat{X}^{\text{Vision}})$$
$$G_t = \text{MCA}(\hat{X}^{\text{time}}, \hat{X}^{\text{text}})$$
$$X^{\text{fuse}} = \hat{X}^{\text{time}} + G_v + G_t$$

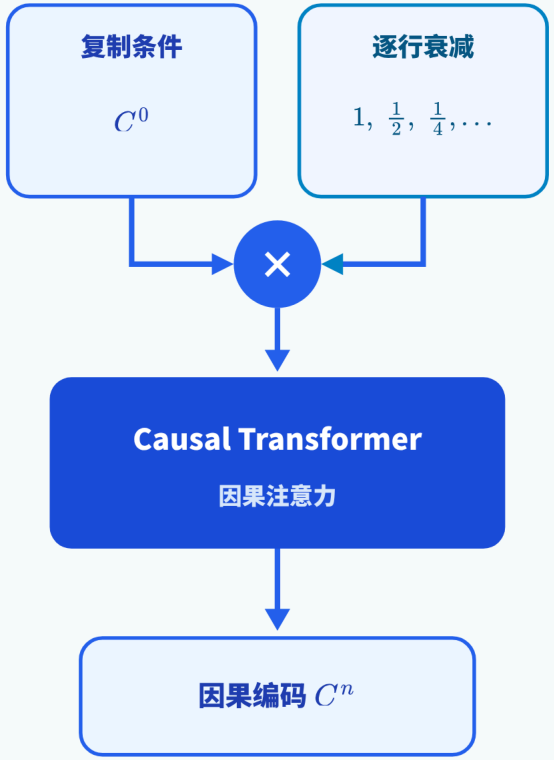
未来条件解码

STEP 1 复制最后一个融合特征



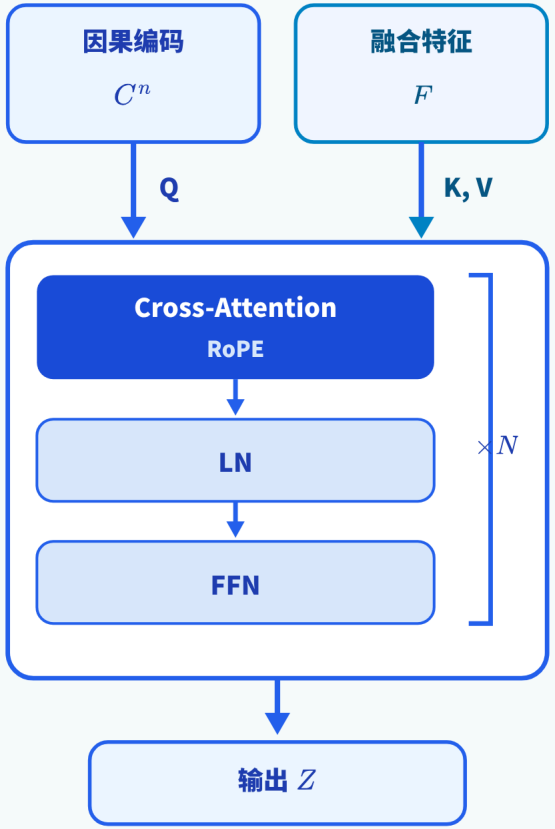
$$F = X^{\text{fuse}}, \quad f_{\text{last}} = F_{\text{last}}$$
$$C^0 = \text{Repeat}(f_{\text{last}}, N)$$

STEP 2 衰减条件与因果注意力



$$w_i = 2^{-(i-1)}, \quad \bar{C}_i^0 = w_i C_i^0$$
$$C^n = \text{CausalTransformer}(\bar{C}^0)$$

STEP 3 N 层 RoPE 交叉注意力

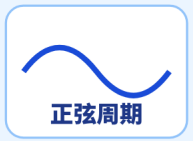


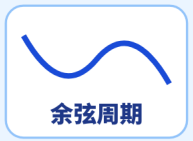
$$Q = C^n, \quad K = F, \quad V = F$$
$$Z = \text{CrossAttentionStack}_N(C^n, F)$$

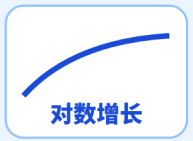
先验原型检索与流匹配生成

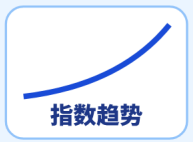
STEP 1 多尺度先验原型库软检索

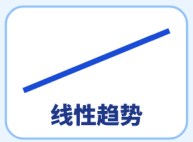
Prototype Bank (周期基 + 趋势基原型库)















PrototypeRetriever

加权合成先验原型 $\tilde{\mathcal{P}}$

$$\alpha_{\text{proto}} = \text{Softmax}(Q_{\text{mm}} B_{\text{bank}}^{\top} / \sqrt{D})$$

$$\tilde{\mathcal{P}} = \alpha_{\text{proto}}\, B_{\text{bank}}$$

$B_{\text{bank}} \in \mathbb{R}^{M_b \times D} \rightarrow \tilde{\mathcal{P}} \in \mathbb{R}^{N_p \times D}$

STEP 2 原型锚定条件流匹配速度场学习

从合成原型出发的最优传输直线轨迹



恒定目标速度向量 • 零曲率最短路径

输入状态与条件

Z_τ, X^{cond}

Flow-Matching Net

$v_\theta(Z_\tau, \tau \mid X^{\text{cond}})$

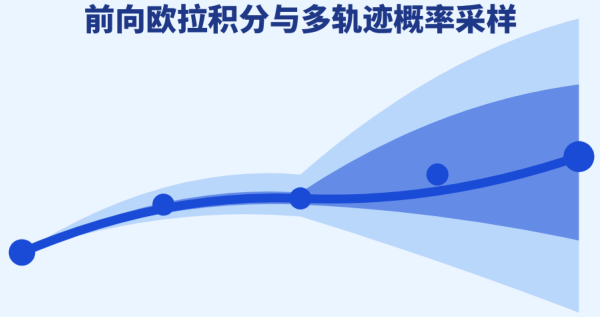
$$Z_\tau = (1-\tau)(\tilde{\mathcal{P}} + \sigma_0 \epsilon) + \tau Z_1$$

$$\mathcal{L}_{\text{FM}} = \mathbb{E}\left\| v_\theta(Z_\tau, \tau \mid X^{\text{cond}}) - (Z_1 - Z_0) \right\|_2^2$$

$v_\theta \in \mathbb{R}^{N_p \times D}$

STEP 3 前向欧拉积分与逆归一化还原

前向欧拉积分与多轨迹概率采样



多次采样噪声生成未来概率置信区间

拼接未来块 + RevIN 逆归一化还原

还原真实物理量纲输出预测 $\hat{Y} \in \mathbb{R}^{H_f}$

$$Z_{\tau+\Delta\tau} = Z_\tau + \Delta\tau\, v_\theta(Z_\tau, \tau \mid X^{\text{cond}})$$

$$\hat{Y} = \text{RevIN}^{-1}(Z_1 W_{\text{out}})$$

$\mathbb{R}^{N_p \times D} \rightarrow \hat{Y} \in \mathbb{R}^{H_f}$

批判 1

01 存在主张缺乏证据

关于流匹配初始分布，论文提出：

> However, current methods still set the initial distributions as Standard Gaussians, which neglects the capability of Flow Matching to work like a stochastic interpolant.
>
> **Obviously, constructing appropriate prototype as the inital starting point can enhance the intuitiveness and stability of Flow Matching.**

> 然而，现有方法仍将初始分布设为标准高斯分布，忽视了流匹配作为随机插值器的能力。
>
> **显然，构造合适的原型作为初始起点，可以增强流匹配的直观性和稳定性。**

02 模块设计过于复杂

模块数量多、设计过于刻意，整体结构复杂。

03 合成语料无法核验

数据集未公开，语料由 AI 合成，无法验证其合理性。

> To generate the textual descriptions for the cross-domain multimodal time series corpus, we provide domain descriptions and raw time series for GPT-4, encouraging it generate the descriptions of data characteristics, which are only used for pretraining Aurora.

据 OpenReview，人工监督了 5%，其余 95% 的内容生成没有人工监督。

评价：这种做法很容易偏离真实情况。

批判 2

04 没有任何消融实验

没有任何消融实验，无法证明任何一个模块是否真的有效，尤其是在模块套模块的情况下。

05 文本语义作用存疑

独立研究在 Time-MMD 上干预文本内容：

实验：将原文替换为空、常量、打乱、跨域或含未来值的文本，比较 Aurora、TaTS 和 MM-TSFlib 的预测。

结果：三种模型的平均 MSE 变化均小于 0.5%；最大变化为 +0.16%。Aurora 的变化在 ±0.05% 内。

分析：TaTS 和 MM-TSFlib 的增益受到同路径数值先验影响；Aurora 的文本路径在该实验中对内容不敏感。

启用与禁用文本路径的对比，不能单独证明模型利用了文本语义。

来源：arXiv:2608.22321，§4–5

06 效果来源有待验证

猜测效果改善可能来自数据泄露，或 Prototype Bank 引入的先验。

Prototype 的部分设计与 TimeBase（ICML 2025）中的方法类似：

TimeBase 通过时间序列分解与矩阵学习得到潜在高频基，并用损失项抑制高频基之间过于相似。

序列可表示为高频基的线性组合与低频成分之和。